

BEN HOWARD

Opelika, AL | 469-601-5589 | benjaminhoward03@gmail.com

EDUCATION

Auburn University, Samuel Ginn College of Engineering
Bachelor of Electrical Engineering
GPA: 3.06 / 4.00

Auburn, AL
Spring 2026

HONORS

IEEE Eta Kappa Nu – Xi Chapter
Dean's List
Outstanding CSSE Project
Outstanding Project

Spring 2024 – Spring 2026
Fall 2022, Spring 2023, Fall 2023
Spring 2026 COE Capstone Showcase
Spring 2026 Auburn ECE Capstone Showcase

TECHNICAL SKILLS

Python, MATLAB, C, LTSpice, Digital Signal Processing, MULTISIM, KiCAD, Microsoft Office Suite

EXPERIENCE

Auburn University MRI Research Center, Auburn, AL

May-July 2025

Undergraduate Research Assistant

- Learning and applying data and image processing techniques for use in MRI research.
- Using MATLAB and Python tools for data and signal processing and muscle segmentation.

Albers Aerospace, McKinney, TX

May-July 2024

Systems Engineering Intern

- Applied systems engineering principles to design a prototype hybrid-electric UAV from requirements through to validation.
- Made design decisions regarding aircraft weight, mechanical and electrical power requirement for power generation and thrust.
- Designed and implemented a power generation, charging, and distribution system for the entire aircraft.

HighPrep Tutoring, Auburn, AL

Tutor

- Tutoring many pre-engineering math and science classes such as Calculus I & II and Physics I & II

PROJECTS

Capstone Design: Sphero Bolt+ Golf Informatics

- Developed a Python API to control a Sphero BOLT+ and capture putting surface metrics.
- Built a GUI app integrating a decision tree model to predict Stimp ratings from sensor data.

Automatic Highlight Detection and Captioning

- Utilized a Boundary Matching Network (BMN) to automatically detect start and end of video highlights.
- Implemented LLaVA to identify on-screen actions translating them into LLM tokens for text captions.

MATLAB-Based FMCW Radar Simulation

- 2D simulation of EM wave transmission, propagation, reflection, and transmission
- Processed the time shifted and doppler shifted waves to calculate object distance and velocity

LEADERSHIP

All Christ Church Leadership Development Program

Fall 2025 – Spring 2026

American Institute of Chemical Engineers (AIChE) – Car Team, Auburn University

Fall 2023 – Fall 2026

Technical Team Lead

IEEE Eta Kappa Nu Honor Society – Xi Chapter

Spring 2024 – Spring 2026

Outreach Coordinator, Treasurer